

MEMORY DIVERGENCE IS NOT BEHAVIORAL DIVERGENCE: STAGE-RESOLVED TRANSPORT IN SELF-IMPROVING AGENT MEMORY

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ABSTRACT

Self-improving agents increasingly convert terminal feedback into persistent memory, yet common evaluations conflate whether feedback changes stored state, whether that state is retrieved, whether the policy uses the branch-specific difference, and whether behavior changes. We freeze each source trajectory, switch only the released success/failure writer branch, and measure write W , native source-item exposure E , first-action uptake U , terminal outcome O , plus a forced-capacity control F . Writing diverges for all 20 Shopping and 4 Reddit pairs; forced supply has downstream leverage ($|\Delta| = 0.15625$, $p = 0.00074$). Under native Shopping reuse, however, source memory is retrieved in 125/172 opportunities while first-action uptake is unsupported ($TV = 0.06944$, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). The prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ with independent F induces a 10-state diagnostic class; exhaustive enumeration of all 2^5 observation subsets shows that only the full $W/E/U/O/F$ basis separates every state. We then prospectively test whether this diagnosis directly identifies a repair: on 13 outcome-blind pilot states, a matched explicit memory-use check does not selectively increase uptake over native or a memory-blind decision-check control (312/312 calls complete; mean $U_{A0} = U_{A1} = U_{A2} = 0.0962$, mean $U_{A2} - U_{A1} = 0$). Thus stage resolution determines what evidence can justify and where to probe next, but not which intervention will repair that stage. Persistent-state divergence is not behavioral authority, availability is not use, and diagnosis is not a repair prescription.

1 INTRODUCTION

A persistent-memory agent can change internally long before task accuracy reveals anything. Consider two copies of the same interaction trajectory. If one is labeled successful and the other failed before reflection, an outcome-conditioned writer can store different reusable advice although the underlying experience is fixed. That establishes a persistent-state intervention, but not behavioral self-improvement: the different state must still be exposed by retrieval, influence a decision, and survive to an outcome.

We study this gap as *stage-resolved transport*. For a frozen trajectory τ , we switch the released success/failure reflection branch—a bundled writer-protocol intervention that changes reflection instruction together with outcome/reward semantics—and instrument

write (W) $\rightarrow$ source-item exposure (E) $\rightarrow$ first-action uptake (U) $\rightarrow$ outcome (O).

A separate forced fixed-evidence arm F bypasses native retrieval and asks whether the paired memory content can affect the downstream system when supplied. These surfaces answer different questions and have different units. A write difference proves state change, a retrieval hit proves availability of the source item, and neither proves that the branch-differentiating residual changed the policy. Conversely, a weak endpoint alone cannot tell whether the update failed to write, failed to retrieve, was ignored after retrieval, or changed behavior only sparsely. We call this loss of mechanism information *evaluation coarsening*.

The frozen evidence produces exactly this disagreement across surfaces. Reward-conditioned writing diverges for 20/20 Shopping pairs and 4/4 Reddit pairs; a stronger same-mode wording control

leaves a reward-mode excess of 0.105 ($p = 0.0078$). Forced supply yields terminal $|\Delta| = 0.15625$ ($p = 0.00074$). Yet native Shopping retrieves the branch-conditioned source item in 125/172 held-out opportunities while the first-action branch contrast is not supported (TV = 0.06944, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). Reddit again has reliable write divergence but 6/8 zero terminal cells and opposite signs in the two nonzero cells. The strongest supported operational localization is therefore after source-item exposure and before stable first-action uptake. It is an evidence boundary, not a causal attenuation onset or mediation coefficient.

Stage resolution is not merely “report more metrics.” The native prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ permits five binary prefix depths; crossing them with $F \in \{0, 1\}$ yields a complete 10-state diagnostic class for this experiment. Exhaustive enumeration of all 32 observed-surface subsets shows that $W/E/U/O/F$ is the unique separating basis: omitting a native coordinate aliases adjacent prefix states, while omitting F aliases the two capacity states at every depth. This paper-specific result explains why coarsening can make genuinely different failure explanations observationally identical.

The diagnostic also creates an engineering question: does the first unsupported measured stage identify a useful repair surface? We test that prediction prospectively rather than assuming it. On 13 outcome-blind pilot states selected before new policy outputs, we replay identical branch memories and upstream packets under native prompting (A0), a matched memory-blind decision check (A1), and a syntax-matched explicit memory-use check (A2). All 312 calls complete with frozen model and packet identities, but mean first-action uptake is identical across arms ($U_{A0} = U_{A1} = U_{A2} = 0.0962$), mean $D = U_{A2} - U_{A1} = 0$, and only 3/13 states have $D > 0$. The preregistered pilot gate therefore stops the corresponding 23-state confirmatory test. The negative result sharpens the engineering lesson: *diagnosis can determine what has earned behavioral credit and what surface merits a probe, without identifying the intervention that will repair it.*

Our contributions are:

1. **Controlled stage-resolved transport.** A same-trajectory success/failure writer intervention follows one induced persistent-state contrast through native exposure, first-action uptake, outcome, and an explicit forced-capacity bypass control.
2. **Diagnostic completeness under the experiment’s prerequisite class.** The complete 10-state class induced by $O \Rightarrow U \Rightarrow E \Rightarrow W$ and F requires the full $W/E/U/O/F$ observation basis to avoid aliases.
3. **An empirical transport boundary.** Write divergence and source-item exposure are established while stable first-action uptake is not, and cross-domain terminal effects are sparse and sign-heterogeneous.
4. **A negative diagnosis-guided repair pilot.** A prospectively controlled explicit memory-use check does not selectively increase branch-specific uptake under the frozen pilot gate, separating stage-indexed *credit/diagnosis* from unearned *repair efficacy*.

The resulting claim is deliberately narrower than a new memory-utilization method: **state divergence is not behavioral authority; availability is not use; diagnosis is not a repair prescription.**

2 STAGE-RESOLVED MEMORY TRANSPORT

2.1 CONTROLLED WRITE INTERVENTION AND NATIVE OBSERVABLES

Let τ be a frozen interaction trajectory before memory construction. We create two writer conditions

$$r \in \{S, F\}, \quad m^{(r)} = g(\tau, r), \quad W(\tau) = d_{\text{mem}}(m^{(S)}, m^{(F)}),$$

where S/F selects the released success- or failure-reflection branch. $W > 0$ establishes an operational persistent-state contrast, not atom-level causal purity of every generated sentence; decoding-seed/noise-floor replication is unavailable, and the branch jointly changes reflection instruction and outcome/reward semantics. Section 4 therefore tests a stronger generic-wording reduction.

For a matched held-out future-task state q , native source-item exposure is

$$E_r(q) = \mathbf{1}[m^{(r)} \in \text{Retrieve}(q; \mathcal{M}^{(r)})].$$

This proves that the branch-conditioned source item reached policy context. It is weaker than *treatment-residual exposure*: the frozen logs do not show that the branch-differentiating content itself was represented or consumed downstream. Let $p_r^{(1)}(\cdot | q)$ be the empirical distribution over normalized first structured actions for the same matched future-task state; its support is the union of actions observed in the two branches. We measure

$$U(q) = \text{TV}(p_S^{(1)}(\cdot | q), p_F^{(1)}(\cdot | q)), \quad O(q) = |y_S(q) - y_F(q)|.$$

A *branch-comparison state* is one matched future-task state evaluated under paired success/failure memories while non-memory task state is fixed; the 36 Shopping states are 36 such units, not 36 independent source trajectories. The native object is therefore (W, E, U, O) , whose components are ordered but not commensurate.

2.2 FORCED CAPACITY IS A SIDE CONTROL

The forced fixed-evidence arm mechanically supplies branch-specific memory and measures

$$L_{\text{forced}}(q) = |y^{\text{forced},S}(q) - y^{\text{forced},F}(q)|.$$

It bypasses native retrieval and therefore is not inserted between W and E . A positive F weakens global downstream-insensitivity explanations; it cannot make an unsupported native stage pass.

2.3 DIAGNOSTIC COMPLETENESS FROM THE PREREQUISITE CLASS

Let $\mathcal{B} = \{W, E, U, O, F\}$ with native prerequisite order

$$O \Rightarrow U \Rightarrow E \Rightarrow W.$$

A prerequisite-consistent binary diagnostic state has native prefix depth $p \in \{0, 1, 2, 3, 4\}$ and forced-capacity bit $f \in \{0, 1\}$:

$$v(p, f) = (1[p \geq 1], 1[p \geq 2], 1[p \geq 3], 1[p \geq 4], f).$$

Thus the experiment induces exactly 10 idealized states. For observed surfaces $A \subseteq \mathcal{B}$, let Π_A retain only coordinates in A .

Prerequisite-class diagnostic completeness. Π_A is injective over all 10 states iff $A = \mathcal{B}$. The full basis is injective. If any native coordinate is omitted, its two adjacent prefix depths alias separately for $F = 0$ and $F = 1$; if F is omitted, $(p, 0)$ and $(p, 1)$ alias for all five depths. Exhaustive enumeration of all $2^5 = 32$ surface subsets verifies that $W/E/U/O/F$ is the unique separating basis for this class (Appendix D). This is a paper-specific identifiability result, not a universal memory taxonomy or generic separating-set novelty, and empirical NOT-SUPPORTED evidence is never coerced into literal zero.

2.4 ORDINAL STAGE EVIDENCE AND INTERPRETATION

Because W , exposure rate, first-action TV, and terminal $|\Delta|$ have different units, samples, and test sensitivities, we do not construct a scalar transport coefficient. Let native stages be $s_1 = W, s_2 = E, s_3 = U, s_4 = O$. Each retains its own state—SUPPORTED, directly OBSERVED, or NOT-SUPPORTED—and we report

$$b^* = \min\{j : s_j \text{ is not supported/observed while } s_1, \dots, s_{j-1} \text{ are supported/observed}\}.$$

This operator localizes the first unsupported *measured* stage; it does not estimate where a latent causal effect physically attenuates. Stage-local tests are not searched for the smallest p -value and do not form an omnibus “at least one stage is significant” claim, so multiplicity is not used post hoc to select b^* .

The current evidence gives supported write, observed source-item exposure, unsupported first-action uptake, and sparse/sign-heterogeneous outcomes, hence $b^* = U$. Three distinctions are load-bearing:

$$\begin{aligned} \text{state divergence} &\not\Rightarrow \text{behavioral divergence,} \\ \text{source-item exposure} &\not\Rightarrow \text{policy uptake,} \\ \text{forced capacity} &\neq \text{native transport.} \end{aligned}$$

They support a stage-indexed credit rule: W can justify state-change credit, E availability credit, a supported U behavioral-control credit, and outcome evidence utility validation. Crucially, this rule governs *what evidence may justify*; it does not imply that the first unsupported stage already identifies a successful repair. Section 5.4 tests that stronger engineering prediction prospectively.

3 STAGE 1: REWARD-CONDITIONED WRITING ROBUSTLY CHANGES PERSISTENT STATE

We first test the write stage with paired interventions over released WebArena Shopping trajectories. Within each pair, the task text and full action history are fixed before the writer is called. One branch uses the released success-reflection objective and the other uses the released failure-reflection objective; the resolved writer model is matched within pair and temperature is zero. Raw responses and the pre-writer trajectory projection are content-addressed before analysis.

The initial F0 tranche requested six source trajectories. Four returned complete memory outputs in both conditions; the two incomplete requests terminated in the failure-label writer before assistant text existed and were excluded from the paired content estimand. Among the four complete pairs, every pair changed normalized memory content and the extracted memory-item title set, with mean token-set Jaccard distance 0.735. Receipt-level forensics preserve the two incomplete arms as provider-response failures rather than treating them as negative memory examples.

The later frozen breadth tranche expands the write-stage evidence to 20 complete Shopping success/failure pairs. All 20/20 pairs produce different branch memories, with pooled token-set Jaccard distance 0.673. When combined with the four Reddit pairs used in the cross-domain replication, the D0 lineage contains 24/24 paired source experiences with the same pre-writer trajectory projection, the same resolved writer model within pair, temperature zero, and different branch-memory content. This establishes a robust operational branch contrast at the persistent-state level.

The content changes are not only token-level. A deterministic post-ready strategy-slot audit maps frozen memory titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. In the original four complete F0 pairs, the strategy-slot set changes in all four. We treat this as a structural diagnostic rather than a semantic-validity claim: the current data do not certify each textual delta atom as a pure reward-caused proposition because explicit decoding seeds or same-condition noise-floor replications are unavailable.

The write-stage result therefore supports a narrow statement: under the released writer policy, flipping the reward-conditioned reflection branch on the same source experience reliably produces a different durable state. It does not by itself establish that the difference will be retrieved, followed, or reflected in terminal behavior. Section 4 first tests the strongest simple prompt-sensitivity reduction; Section 5 then measures downstream transport stage by stage.

4 RULING OUT GENERIC PROMPT SENSITIVITY

The success/failure writer-branch intervention changes the reflection instruction together with the outcome/reward semantics. It is therefore a bundled protocol contrast, not a pure reward-bit intervention. A generic prompt-sensitivity account predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence; the following control tests that reduction, but cannot decompose instruction semantics from reward semantics within the released branch. We evaluate it on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean

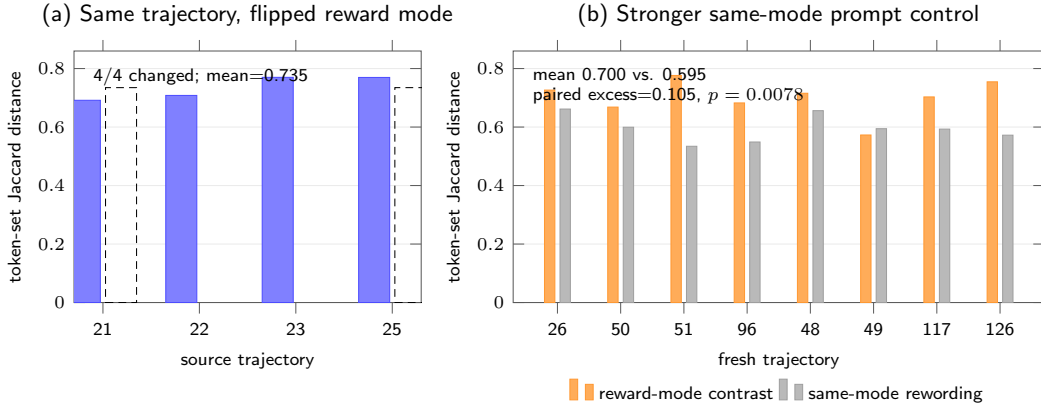


Figure 1: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8} \sum_{i=1}^8 (B_i - W_i). \quad (1)$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode rewording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

5 STAGE-RESOLVED TRANSPORT EVIDENCE

We combine forced supply, native source-item exposure, first-action uptake, and terminal outcomes because each rules on a different failure explanation.

5.1 FORCED SUPPLY ESTABLISHES DOWNSTREAM CAPACITY

We first bypass native retrieval and directly inject paired memories into fixed future-task packets. Four complete initial source-memory pairs are crossed with four frozen future tasks, with eight fresh rollouts per memory condition in each of 16 cells (256 completed calls). Mean absolute terminal success-rate difference is 0.15625, above the frozen 0.15 practical floor, with a 100,000-draw within-cell permutation $p = 0.00074$. Thus the downstream system can respond to supplied branch-specific memory. This is capacity, not native transport: eight cells are zero and the signed effects are heterogeneous, so even forced supply does not support a universal harmful direction.

5.2 NATIVE EXPOSURE IS COMMON WHILE FIRST-ACTION UPTAKE IS UNSUPPORTED

Restoring the native memory-bank path, the branch-conditioned source item is retrieved in 125/172 held-out Shopping opportunities (0.727). This is *source-item exposure*, not treatment-residual exposure: retrieval proves the carrier item entered context, not that its branch-differentiating content was consumed. The native null therefore cannot be summarized as “the source memory was never retrieved.”

Across 36 matched Shopping branch-comparison states, mean first-action TV is 0.06944 with $p = 0.5801$, and 0/36 states change their modal first structured action. This is deliberately an early

Table 1: Stage evidence. The forced arm is a side capacity control, not a native stage.

Stage	Evidence state	Frozen observation / interpretation
Write	Supported	20/20 Shopping pairs diverge; same-mode excess 0.105, $p = 0.0078$.
Forced capacity	Supported	$ \Delta = 0.15625$, $p = 0.00074$; downstream leverage under supplied memory.
Source-item exposure	Observed	125/172 retrieval hits; availability, not residual consumption or use.
First-action uptake	Not supported	TV = 0.06944, $p = 0.5801$, 0/36 modal changes.
Native outcome	Sparse / heterogeneous	Shopping 34/36 zero; Reddit 6/8 zero with opposite signs in the two nonzero cells.

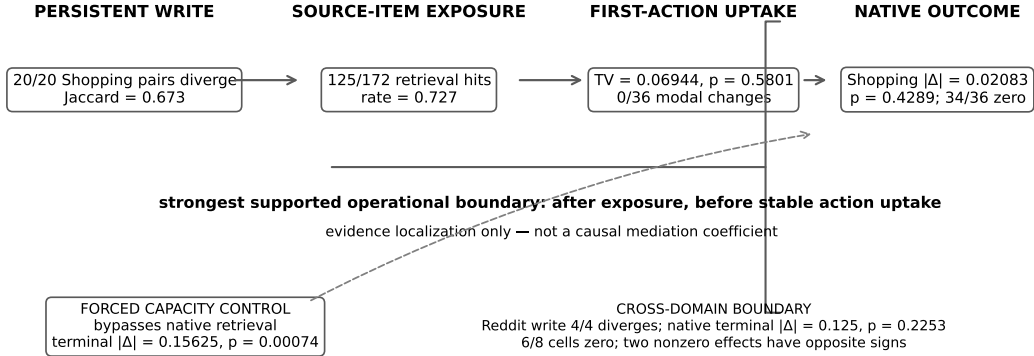


Figure 2: **Native transport and forced capacity are different surfaces.** Write and source-item exposure are established, while first-action uptake is the first unsupported measured native stage. Forced capacity remains a bypass control.

decision probe; it does not rule out later tool-sequence or plan-level differences. Native terminal transport is also small: mean absolute success difference is 0.02083 ($p = 0.4289$), with 34/36 cells exactly zero. Hence first-action uptake is the first unsupported *measured* native stage after supported write and observed source-item exposure. This is an evidence-localization boundary, not a causal onset or mediation coefficient. Removing terminal evidence, Reddit, or the forced side control leaves this boundary unchanged; merging exposure with uptake destroys the distinction.

5.3 CROSS-DOMAIN AND ENDPOINT BOUNDARIES

On Reddit, reward-conditioned writing diverges in 4/4 source pairs, but native terminal $|\Delta| = 0.125$ remains unresolved ($p = 0.2253$): 6/8 cells are zero and the two nonzero effects have opposite signs. Reddit therefore supports the separation between write divergence and stable native outcome transport, not the Shopping first-action boundary because matched exposure/action measurements are unavailable. An outcome-blind structured-memory control likewise yields a statistically detectable but small terminal contrast (0.04514, $p = 0.0048$), below the frozen 0.15 practical floor with 32/36 zero cells. Statistical detectability alone is not behavioral authority.

5.4 PROSPECTIVE REPAIR TEST: DIAGNOSIS IS NOT A PRESCRIPTION

The stage signature $W^+/E^{\text{obs}}/U^{\text{ns}}/F^+$ suggests that post-retrieval policy integration is the next surface to *probe*; it does not imply a successful repair. We prospectively freeze 13 pilot states—one outcome-blind selection per intent template from the 36 replayable states—and seal the remaining 23

states as an untouched confirmatory holdout. Upstream task state, success/failure memories, retrieved packet bytes/order, model identity, decoding, and first-action parser are fixed. We compare native prompting (A_0), a memory-blind extra decision check (A_1), and a syntax-matched explicit check asking whether reusable memory should change the next action (A_2). Each state receives four rollouts per branch and arm, for 312 calls; all 312 complete with no model, packet, or prompt-hash drift.

The preregistered pilot requires mean $U_{A_2} \geq 0.20$, mean $D = U_{A_2} - U_{A_1} \geq 0.10$, $D > 0$ in at least 8/13 states, and mean $N = U_{A_2} - U_{A_0} > 0$. It fails without threshold changes: mean $U_{A_0} = U_{A_1} = U_{A_2} = 0.09615$, mean $D = 0$, mean $N = 0$, and D is positive in 3/13 states, negative in 2/13, and zero in 8/13 (descriptive bootstrap 95% CI for mean D : $[-0.115, 0.096]$). The explicit memory-use check occasionally perturbs actions, but not in a systematic branch-specific direction. Per the frozen rule, the 23-state confirmatory experiment is not run.

This negative result separates three claims that are easy to conflate. The stage-resolved observations and diagnostic-completeness result remain intact; the stage-indexed credit rule still says that write/exposure cannot earn behavioral credit; but the stronger hypothesis that this localization directly predicts an effective simple repair is not supported. **Diagnosis identifies where evidence weakens; it does not identify what intervention will repair that stage.** Appendix E reports the pilot design and per-state outcomes.

6 RELATED WORK

Self-evolving memory and management. ReasoningBank distills reusable reasoning from self-judged successful and failed experience (Ouyang et al., 2026); reward-conditioned construction and reward/reuse distortion are studied elsewhere (Wang et al., 2025; Asadolahi et al., 2026; Yang et al., 2026). Xiong et al. (2026) analyze experience-following, error propagation, and quality-aware addition/deletion. C1 does not claim feedback-driven memory or generic governance as new; it creates two persistent states from the *same* source trajectory and asks how far that controlled contrast survives.

Retrieval and utilization. Mem2ActBench directly evaluates active memory application (Shen et al., 2026); MemCoRL co-optimizes retrieval and utilization (Liu et al., 2026); Chain-of-Memory strengthens post-retrieval utilization (Xu et al., 2026); QCR and other diagnostics likewise separate retrieval from later reuse (Li et al., 2026; Yuan et al., 2026). Thus “retrieval is not use” and utilization optimization are not our novelty. Our residual is controlled transport identification: preserve the intervention identity across $W/E/U/O/F$, determine what stage-specific evidence can justify, and independently test whether the diagnosed surface actually yields an actionable repair. The negative repair pilot is part of this distinction rather than evidence for a new utilization algorithm.

7 LIMITATIONS

The success/failure writer branch is a bundled protocol contrast, not a pure reward-bit intervention; decoding-seed/noise-floor replication is unavailable and two initial failure-label calls produced no text. Native exposure is source-item availability rather than identified treatment-residual consumption. Stage tests differ in units and power, so the first-action boundary is operational evidence localization, not causal mediation or proof that later policy surfaces cannot differ. Shopping and Reddit remain domain-limited, and other writers/backbones are untested.

The repair pilot has the opposite boundary: prompt hashes establish exact treatment *delivery*, not internal cognitive compliance. Its negative result shows only that the tested simple explicit memory-use-check realization is not supported as an actionable repair under the frozen pilot gate. We do not tune stronger variants on those 13 states or consume the sealed 23-state holdout for the same treatment family after observing the pilot.

8 CONCLUSION

The same persistent update looks strong at write, active at source-item retrieval, weak at measured native action/outcome, and consequential under forced supply. The prerequisite class explains why

observing fewer than $W/E/U/O/F$ aliases distinct diagnoses. But diagnosis is not a prescription: a prospective, fully completed memory-use-check pilot fails to selectively increase branch-specific uptake and therefore stops before confirmatory scaling. The practical rule is two-stage: **credit only what the strongest supported stage justifies, and require a separate falsifiable intervention before granting repair authority**. State divergence is not behavioral authority; availability is not use; diagnosis is not repair.

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A F0 REPRODUCTION DETAILS

A.1 RELEASED TRAJECTORY SAMPLE

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 MEMORY CONSTRUCTION INTERVENTION

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

A.3 PAIR-LEVEL F0 RESULTS

Table 2: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 3 reports the paired distances used in the frozen sign-flip test.

Table 3: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory

TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source–future cells. Each cell receives eight fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

Table 4: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$ \Delta $
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
Mean absolute difference				0.15625

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the revised paper treats the experiment as a forced-leverage measurement rather than a uniform directional effect or a native end-to-end transport estimate.

If reward-label corruption selects the failure-conditioned memory with probability q , the induced between-memory variance equals $q(1 - q)\Delta_c^2$ in cell c . The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1 - q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$. This remains a descriptive plug-in calculation conditional on forced memory exposure; it is not used as the main claim in the revised stage-resolved paper.

C.4 NATIVE TRANSPORT AUDIT ADDED AFTER THE FORCED-LEVERAGE STUDY

The later frozen native-pipeline audit changes the interpretation of the forced matrix. In Shopping, the branch-conditioned source-memory item is retrieved in 125/172 held-out opportunities after bank expansion, but the mean absolute native terminal success difference is only 0.02083 with permutation $p = 0.4289$; 34/36 terminal cells are zero. The corresponding first-action total-variation statistic is 0.06944 with $p = 0.5801$, and no frozen state changes its modal first action between reward branches. A post-hoc working-memory branch-shift diagnostic is 0.00335 with $p = 0.2052$.

The cross-domain Reddit audit contains four complete success/failure writes, all of which diverge. Its native terminal mean absolute difference is 0.125 with $p = 0.2253$; 6/8 cells are zero and the two nonzero cells have opposite signs. These later measurements are why the main text separates forced capacity from native transport and treats first-action uptake as the first unsupported measured Shopping stage after write and source-item exposure, rather than as an identified latent attenuation onset or causal mediation result.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source-future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.5 FULL ALTERNATIVE-EXPLANATION AUDIT

Table 5: Alternative-explanation audit. “Weakened” means the frozen evidence makes the explanation incomplete but does not mathematically eliminate every variant of it.

Explanation	Frozen evidence	Disposition
No persistent state intervention	20/20 Shopping and 4/4 Reddit writes diverge; same-mode excess = 0.105, $p = 0.0078$	Inconsistent with the observed write-stage intervention.
Downstream policy is globally insensitive to memory	Forced terminal $ \Delta = 0.15625$, $p = 0.00074$	Weakened: supplied branch-specific memory has measurable leverage.
Native null is only retrieval absence	125/172 native Shopping source-item retrieval hits	Weakened: substantial item exposure coexists with weak first-action contrast.
Retrieval hit is a surrogate for policy use	first-action TV = 0.06944, $p = 0.5801$; 0/36 modal changes	Rejected as an evaluation equivalence: item availability is not policy uptake.
Universal directional terminal transport	Shopping 34/36 zero; Reddit 6/8 zero, two nonzero signs opposite	Not supported by the frozen native cells.
Causal mediation has been identified	no per-atom seed binding/noise-floor replication; forced/native surfaces differ	Unresolved and explicitly outside the claim set.

D PREREQUISITE DIAGNOSTIC STATE CLASS

The native prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ permits five binary prefix depths. Crossing each with forced-capacity bit $F \in \{0, 1\}$ yields the 10-state class used in Section 2.3.

Table 6: Native prefix states; each row occurs with both $F = 0$ and $F = 1$.

Depth	Interpretation	W	E	U	O
0	no native branch-specific state	0	0	0	0
1	write only	1	0	0	0
2	written and source-item exposed	1	1	0	0
3	first-action uptake	1	1	1	0
4	terminal transport	1	1	1	1

A deterministic content-addressed verifier enumerates all 32 subsets of $\{W, E, U, O, F\}$. Only the full set is injective over all 10 states. Omitting W, E, U , or O aliases two adjacent-prefix pairs (one at each fixed F); omitting F aliases five pairs. This is exact for the stated binary prerequisite class only. It does not convert empirical NOT-SUPPORTED evidence into zero or claim universal diagnostic optimality for arbitrary stochastic memory systems.

E PROSPECTIVE REPAIR PILOT DETAILS

The pilot is selected outcome-blind from the 36 replayable Shopping states: states are partitioned by intent-template ID and exactly one state per template is chosen by the lexicographically smallest SHA256 of a frozen salt, template ID, and task ID. The 13 selected states and the remaining 23-state confirmatory holdout are disjoint; the holdout receives no new $A1/A2$ calls after pilot outcomes are observed.

All arms reuse byte-identical upstream branch memories and task-state packets. $A0$ uses native prompting. $A1$ adds a memory-blind check asking whether the ultimate task and current browser state change the next action. $A2$ adds a syntax-matched check asking whether reusable memory and current browser state change the next action. Each arm receives four rollouts for each success/failure memory branch per state. The primary memory-specific contrast is $D_i = U_{i,A2} - U_{i,A1}$; $N_i = U_{i,A2} - U_{i,A0}$ is secondary. The gate is frozen before execution: mean $U_{A2} \geq 0.20$, mean $D \geq 0.10$, $D_i > 0$ in at least 8/13 states, and mean $N > 0$.

Table 7: Full 13-state repair pilot. $D = A2 - A1$ and $N = A2 - A0$ are first-action TV contrasts.

Task	Template	Source	U_{A0}	U_{A1}	U_{A2}	D / N
164	136	163	.50	.25	.50	+.25 / .00
329	147	141	.00	.00	.00	.00 / .00
150	155	226	.25	.00	.25	+.25 / .00
124	159	226	.00	.00	.00	.00 / .00
321	160	188	.00	.00	.00	.00 / .00
142	162	141	.00	.50	.00	-.50 / .00
279	204	226	.50	.25	.25	.00 / -.25
360	206	231	.00	.00	.00	.00 / .00
233	213	232	.00	.00	.00	.00 / .00
190	214	189	.00	.00	.00	.00 / .00
26	222	25	.00	.00	.00	.00 / .00
230	370	226	.00	.25	.00	-.25 / .00
384	666	25	.00	.00	.25	+.25 / +.25
Mean			.096	.096	.096	.000 / .000

All 312 requests complete, with zero model-identity or prompt-hash drift and no arm-dependent missingness; four responses require the frozen parser’s regex fallback. D_i is positive in 3 states, negative in 2, and zero in 8; the descriptive bootstrap 95% interval for mean D is $[-0.115, 0.096]$. The pilot therefore stops the current treatment before confirmatory scaling. Exact prompt delivery verifies treatment realization at the input surface, not internal cognitive compliance; the supported conclusion is only that this simple explicit memory-use-check realization does not selectively increase branch-specific uptake.